Is Development of Mutual Theory of Mind Essential for Developing Human-Level AI?

Ashok K. Goel ashok.goel@cc.gatech.edu
Design Intelligence Laboratory, Georgia Institute of Technology, 225 North Avenue NW, Atlanta, GA 30332, USA
Qiaosi Wang qiaosiw@andrew.cmu.edu
Human-Computer Interaction Institute (HCII), Carnegie Mellon University, 5000 Forbes Avenue, Pittsburgh, PA 15213, USA

Abstract

Research on AI typically focuses on the cognitive aspects of intelligence such as problem solving and learning. However, human intelligence also involves social processes such as communication and collaboration. Human-human communication engages the use of Theory of Mind, the cognitive ability to attribute mental states to others and to oneself. The use of the Theory of Mind in human communication is bi-directional and iterative, giving rise to a Mutual Theory of Mind that is constantly shaped by communicative responses. However, communication between humans and AI agents at present does not entail the construction of a Mutual Theory of Mind. We posit that the development of a Mutual Theory of Mind between humans and AI agents is essential for developing human-level AI. We also postulate that metacognition in an AI agent may be the key to sustain the Mutual Theory of Mind between a human and the AI agent over time.

Keywords: Human-Level AI, Human-AI Interaction, Theory of Mind, Mutual Theory of Mind, Metacognition.

Section 1: Introduction

Since its inception as a field of study in the 1950s, research and development in artificial intelligence (AI) has focused almost entirely on the cognitive aspects of intelligence such as problem solving, learning, vision, and language processing. See, for example, the four editions of *Artificial Intelligence: A Modern Approach* by Stuart Russell & Peter Norvig (2020) that have helped educate a generation of AI students. The twenty-eight chapters in the fourth edition of the textbook are grouped into seven segments titled "Introduction", "Problem Solving and Search", "Knowledge, Reasoning, and Planning", "Uncertain Knowledge and Reasoning", "Machine Learning", "Communicating, Perceiving, and Acting", and "Conclusions", and even the part on "communicating" is almost exclusively about language processing.

Two paradigms have dominated much of the AI research over the last seventy years (Chandrasekeran & Goel 1988). In the paradigm of Symbolic/Knowledge-Based AI, AI research has focused on the nature of deliberative thought and developed knowledge representations and knowledge-based methods for reasoning and learning. Examples include search over symbolic representations

(Newell & Simon 1976) and ontologies for large-scale knowledge representation (Lenat & Guha 1990; McGuiness & van Harmelen 2004). In the other paradigm of Connectionist/Generative AI (GenAI), AI research has focused on artificial neural networks and developed data-driven methods for associative learning and pattern recognition. Recent examples include deep learning (Goodfellow et al. 2016; LeCun et al 2015) and transformer architectures (Vaswani et al. 2017). Both paradigms have had significant successes on specific tasks, for example, expert systems (Buchanan & Shortliffe 1984) and cognitive tutors (Anderson et al. 1994) in KBAI and conversational interfaces (McTear 2022) and image synthesis (Rombach et al. 2022) in case of GenAI. However, neither paradigm thus far has succeeded in achieving human-level intelligence with all its complexity, efficiency, generality, flexibility, and creativity.

We posit that the development of human-level AI requires advances in computational understanding of the social aspects of intelligence in addition to the cognitive aspects. Human intelligence is fundamentally a socio-cognitive system, not just a cognitive system (Vygotsky 1978). Human intelligence engages not only cognitive processes such as reasoning over knowledge representations and learning of associations between words and images,, but also social processes such as human-human communication, cooperation, and collaboration (Tomasello 2014). Further, it is not that humans first fully develop cognitive abilities and only then start social interactions: instead, social interactions are fundamental to the development of the mental representations and processes.

Computational modeling of social processes requires an understanding of Theory of Mind and Mutual Theory of Mind among other constructs. Theory of Mind (ToM) is the socio-cognitive ability to attribute mental states (goals, plans, knowledge, beliefs, feelings) to others and to oneself (Premack & Woodruff 1978; Bar-Cohen et al. 1985; Wellman 2001). ToM enables human A to explain human B's observed behaviors and make predictions about B's future behaviors; ToM is a basic building block of human-human communication. Mutual Theory of Mind (MToM) is the reciprocal and recursive form of ToM where communicative responses continuously inform attributions of mental states to each other. It pertains to how human A's responses inform human B's theory of A's mind, which shapes B's responses, which in turn shape A's theory of B's mind (Corballis 2007; Eicher et al. 2017; Wang & Goel 2021, 2022). MToM enables alignment of mental states between A and B, including expectations and perceptions; MToM is the basis for human-human communication, cooperation, and collaboration. Given our stance that the development of human-level AI requires advances in understanding the social aspects of intelligence and given that ToM and MToM are the basis of the social processes of communication and collaboration, computational modeling of ToM and MToM seems essential for developing human-level AI.

Section 2: ToM and MToM in Human Intelligence

Let us consider the following sentence:

- S1: The keys are in the drawer.

This sentence is an example of a direct observation of a state in a microworld. There is no ascription of a mental state in this sentence.

Now let us consider a second sentence:

- S2: Chandra thinks the keys are in the drawer.

This sentence ascribes a mental state to Chandra. This is an example of ToM.

Next let us consider a third sentence:

- S3: I know that the keys are in the drawer.

In this sentence, I am ascribing a mental state to myself. This too is an example of ToM.

Next consider a fourth sentence:

- S4: Chandra thinks that Maithilee knows that the keys are in the drawer.

This is an example of a recursive ToM.

Finally, consider a fifth sentence:

- S5: Chandra said to Maithilee: Put the keys back after you return from your drive. Maithilee replied: Okay, but I can't find the keys anywhere. Chandra responded: I thought you knew they are in the drawer.

This is an example of MToM: Both Chandra and Maithilee have a recursive and reciprocal ToM of each other and their ToM of each other is evolving because of the communication between them. Thus, Chandra initially thinks that Maithilee knows that the keys are in the drawer. After Maithilee says that she can't find the keys anywhere, Chandra revises his ToM of Maithilee and tells her that the keys are in the drawer.

We note that since ToM and MToM are theories, they can be, and often, incomplete and/or incorrect.

The Role of ToM

ToM allows humans to infer hidden, non-observable mental states (goals, plans, knowledge, beliefs, feelings) from observable behaviors of other humans as well as oneself.

- *Explaining Behavior:* ToM allows to explain someone's behavior instead of treating them like a black box (e.g., *"Maithilee took the keys because she wanted to go for a drive."*).
- *Predicting Behavior:* ToM allows us to anticipate what someone will do based on their internal state (e.g., *"Maithilee will put the keys back in the drawer once she returns from the drive."*).
- *Detecting False Beliefs:* ToM enables humans to realize that others can hold beliefs that contradict objective reality (e.g., *"Maithilee knows that Chandra might think the keys are in the drawer when actually she took them when she went for a drive.")* This capacity is essential for recognizing mistakes, anticipating poor decisions, and detecting deception.*Empathy:* By modeling another person's emotional state, ToM forms the baseline for empathy, enabling us to respond appropriately to different emotional states such as joy or distress (e.g., *Maithilee recognizes that Chandra may be disappointed if he doesn't find the keys in the drawer.)*

The Role of Mutual ToM

While ToM allows us to understand others as well as ourselves, MToM enables us to adaptively interact with one another by understanding ourselves through the eyes of another human. It enables bidirectional and reflective coordination among humans. Two workers carrying a large box up a staircase is a simple example of a task that requires continuous bidirectional and reflective human-human coordination.

- *Pragmatic Communication:* Human language is ambiguous. We omit many details in our conversations because we know what the listener knows and what the listener understands what we know. Although Grice (1989) did not use the terminology of MToM, he showed that making sense of our conversations often entails recursive thinking. Tomasello (2008) argued that this recursive thinking led to shared intentionality among humans and linked it to cognitive evolution.
- *Social Norms:* MToM allows humans to navigate social roles and shared expectations. We adjust our behavior based on what we know others expect of us in each context. Goffmann (1959) called this impression management.
- *Negotiation:* In competitive as well as cooperative scenarios, human intelligence often relies on recursive thinking (*e.g., "I know that you expect me to choose option A, so I will choose option B"*). This capability forms the backbone of negotiation and diplomacy.
- *Pedagogy:* Effective teaching is based on the teacher's ToM of the learner (Strauss & Ziv 2012). Further, an effective human teacher doesn't just explain facts; they use MToM to evaluate what the learner knows, what the learner doesn't know, and how the learner perceives the teacher's explanation. Effective teachers adjust the content delivery dynamically based on their understanding of the individual learner's cognitive and metacognitive states. In fact, much of this article arose out of our research on AI in education.

Section 3: ToM and MToM in Human-AI Interaction

For the purposes of this article, let us imagine that a year from now, or perhaps a decade or a century from now, humans have developed human-level AI. That is, imagine a world in which a community of AI agents is embedded in the human society, that live, work, and play with humans, and that are comparable to humans in their capability for constructing MToM. What might human-AI interactions in this imaginary world be like?

This ability of AI constructing MToM would represent a major shift from the current state of the art in Human-Computer Interaction (HCI) and Human-Centered AI (HCAI): MToM would treat human-AI interaction as an iterative, bidirectional information-processing loop where attributions of mental states are dynamic and situated.. Current HCI and HCAI research mostly focuses on the static, one-directional representations through one of two directions: (i) *The Human's Mental Model of the AI Agent:* How the user understands the AI agent's capabilities and rules. (ii) *The AI's User Model:* How the AI agent infers the user's goals and preferences. MToM would unify these two perspectives into a co-adaptive feedback loop where both the human and the AI continuously construct, update, and refine models of each other throughout the interaction (Wang et al. 2024).

The above co-adaptive feedback loop driven by MToM would have at least three implications:

- ***Dynamic Communication Repair:*** In natural human dialogue, misunderstandings are inevitable. Humans resolve them using communication repair—adjusting explanations based on what they realize the listener misconstrued (Schegloff et al. 1977). MToM enables AI agents to track not just *what* a user said, but *what the user thinks the agent understands*. If the AI agent detects that a user's prompt is corrective, the agent can actively repair the breakdown by revising its inferences about the user rather than returning incorrect output.
- ***Effective Human-AI Teaming*:** In complex collaborative domains (e.g., medical diagnostics, intelligent tutoring, and human-robot teams), successful teamwork requires shared awareness (Cannon-Bowers et al. 1993; Eicher et al. 2017; Endsley 1995). An AI agent with MToM will evaluate whether the human partner trusts it appropriately in each context. If the agent senses that a user is over-relying on its recommendations, it can adaptively highlight uncertainty or prompt human verification.
- ***Supporting Co-Evolution in Long-Living Interactive Systems:*** Unlike static user interfaces of today, conversational AI agents and digital companions will interact with users over extended periods. MToM offers a blueprint for human-AI co-evolution for long-living human-AI assemblages (Wang et al. 2026). As the user learns the system's strengths and limitations over time, the system can simultaneously update its model of the user's evolving expertise and expectations.

Section 4: Role of Metacognition in Developing Mutual Theory of Mind

The question now becomes how may an AI agent construct a ToM of a human user or sustain a MToM over time? The SOAR cognitive architecture does not have a "ToM module"; instead, it uses the mechanisms of impasse, substates, and working memory to model other agent's mental states (Laird 2012). ACT-R too does not have a ToM module: it uses its goal, memory, and imaginal modules to simulate another agent's mental states (Anderson et al. 2004). Wang & Goel (2022) present a three-stage process for the construction of the MToM. To illustrate their account, let us consider a hypothetical conversation between a learner and an intelligent tutoring system as the context. As Figure 1 illustrates, in the first stage, the tutor *constructs* its understanding of the learner's understanding about the learning concepts based on learning data. For example, the AI tutor may infer that the learner is confused about Concept A and offer some resources to help better understand the concept. In the second stage, the learner *recognizes* the AI tutor's understanding of their learning of the concepts and requests help with both Concept A and Concept B (not just Concept A). In the third stage, the AI tutor *revises* its understanding of the learner's understanding of the learning concepts and offers help for both Concept A and Concept B. Thus, the three-stage process results in the kind of dynamic communication repair we discussed in the previous section.

In Wang & Goel's three-stage process account for construction of MToM, the AI agent begins with an incorrect theory of the human's mind and thus makes a mistake in its assessment of the human; then, the human recognizes the agent's error and communicates this to the agent; and next the agent revises its theory of the human's mind. The question then becomes how do we give this process account

computational form in terms of knowledge, representations, and reasoning so that it can be instantiated in a computer program?

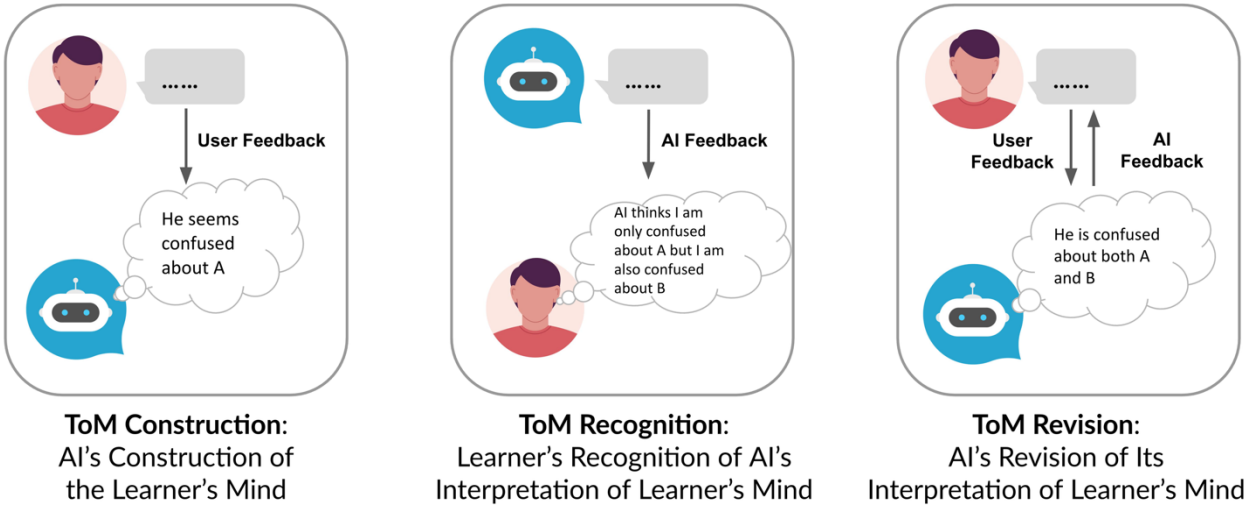


Figure 1. A three-stage process for an AI agent in conversation with a human to construct MToM. This figure has been adapted from Wang & Goel (2022).

We postulate that an AI's metacognition may play a key role in the sustaining the MToM between a human user and the AI over time. Metacognition refers to "cognition about cognition" (Flavell 1979) and plays a central role in self-regulated learning (Brown 1987). Research on AI too has investigated metacognition in the form of "reasoning about reasoning" (Minsky 2005; Cox & Raja 2011). Research on cognitive architectures has investigated metarepresentations or "representations of representations" (Anderson et al. 2004; Laird 2012). Recently in cognitive psychology, Conway et al. (2019) have explored the relationship between metacognition and ToM in the context of representations of whole minds ("mind space") instead of mental states.

In our computational model, when the human recognizes an error made by the AI agent and communicates this to the agent, the AI agent introspects over its theory of its own mind to identify what knowledge or reasoning about the human caused it to make the error, corrects the error and thereby revises its ToM of the human. Thus, this computational model uses metacognition over its model of itself as a fundamental mechanism in sustaining the MToM over time based on communicative responses.

Kim et al. (2026) describe a specific example in detail. One of the common problems in online education for adult learners engaged in reskilling/upskilling is that the class size is large, learners are geographically distributed, and learning is asynchronous. As a result, many online learners feel socially

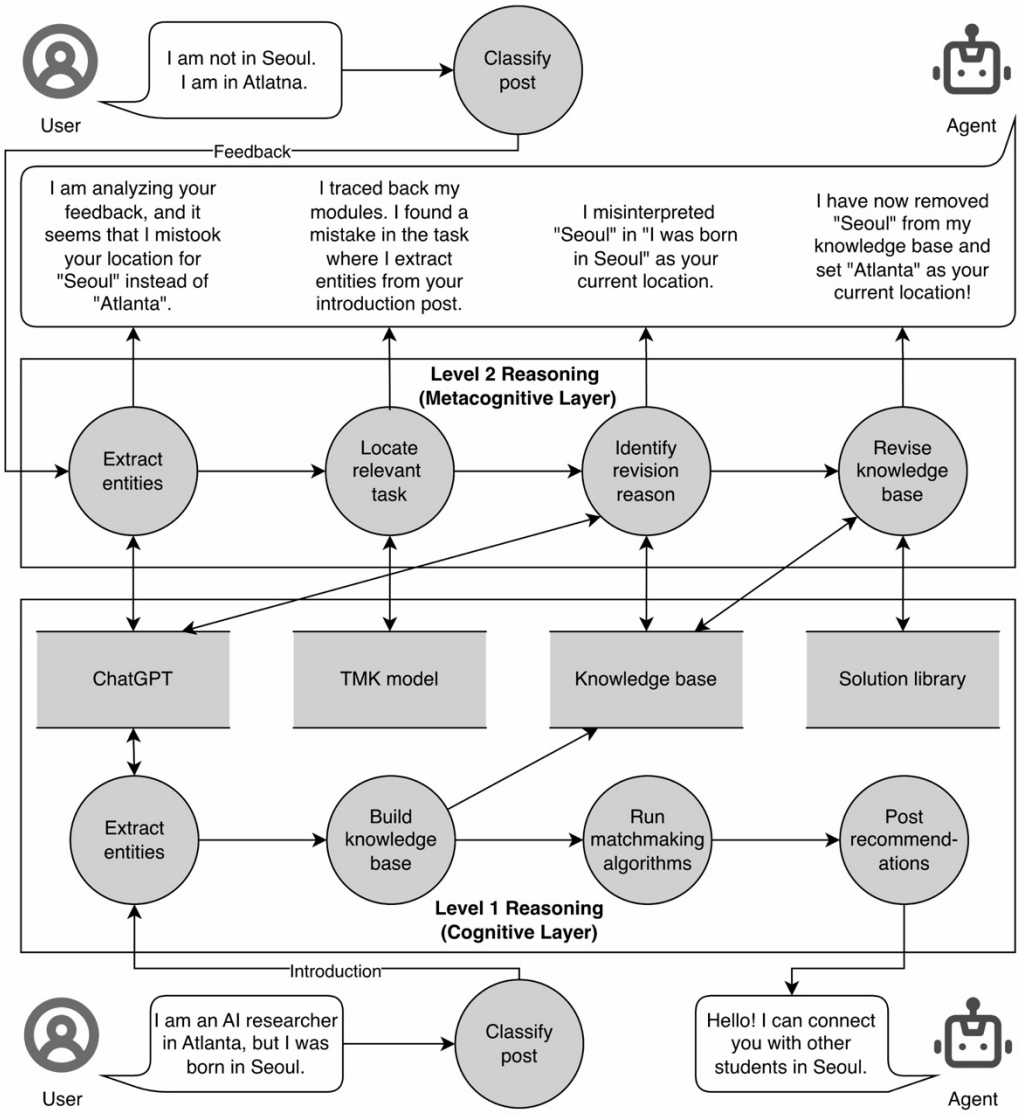


Figure 2. Reasoning in the cognitive and metacognitive layers of the AI agent called SAMI. In the cognitive layer (bottom half of the figure), SAMI may build an incorrect mental model of the human user due to GPT's error. TMK model here refers to Task-Method-Knowledge of the SAMI. When the user corrects SAMI ("I am not in Seoul, I am in Atlanta" in the top left of the figure), then in the metacognitive layer (top half of the figure), SAMI introspects on its theory of its own mind and revises its model of the human. This figure has been adapted from Kim et al. 2026.

isolated which can result in a loss of engagement over time. To address this problem, we developed a social AI agent called SAMI for Social Agent Mediated Interaction (Wang et al. 2020; Kakar et al., 2024) that uses GPT to read learners' posts on the discussion forum for the online class, builds a mental model for each learner in the form of a knowledge graph, performs graph matching, and recommends matches to the learners based on shared locations, backgrounds, and interests. For example, if a learner posts that she lives in Edinburgh and likes hiking, SAMI may recommend

connections with other learners in Edinburgh's time zone as well as other learners who say they enjoy hiking. The bottom half of Figure 2 illustrates SAMI's processing. We have found that the use of SAMI results in an increase in social belonging in online classes as measured by standard Community of Inquiry surveys (Kakar et al. 2024; Wojcik et al., 2025).

However, we also found SAMI sometimes makes mistakes in its recommendations in part because of GPT's hallucinations and partly because of updates in learners' information. As an example of the former, if a learner posts "I am an AI researcher in Atlanta but I was born in Seoul", SAMI may mistakenly infer that the learner lives in Seoul and offer to connect the learner with other learners in the Seoul geographical region. The learner may then recognize SAMI's error and correct SAMI by posting "I am not in Seoul, I am in Atlanta" as indicated in the top left of Figure 3. In addition to the cognitive layer for building a mental model of learners (bottom half of the figure), SAMI contains a metacognitive layer that affords SAMI to introspect on its theory of its own mind illustrated in the top half of the figure. SAMI represents ToM of its own mind in the form of Task-Method-Knowledge models (Goel & Rugaber 2017). Introspection over the TMK model enables SAMI to identify the error in its knowledge graph for the learner. Thus, SAMI corrects its mental model of the learner and matches her appropriately with other learners in the online class. Kim et al. 2026 provide details of this use of metacognition in the construction of MToM.

The above example illustrates one role of metacognition in human-AI interaction that results in a revision of AI's mental model of the human user. However, the interplay between metacognition and MToM remains underexplored in general. We know that humans often build mental models of how AI agents think in analogy to their naïve models of how they themselves think (Wang & Goel 2022). In the other direction, might AI agents develop mental models of how humans think based on their (more accurate and more precise) models of how they themselves think?

Section 5: Conclusions

The path to the development of human-level AI depends on one's perspective on the nature of human intelligence and that of AI itself. A while back, in the heyday of KBAI, one school of thought asserted that building large knowledgebases was the key to human-level AI (Lenat & Guha 1994). Nowadays, in the prime of GenAI, the so-called "scaling hypothesis" asserts that adding more data and more compute is the key to human-level AI (Kaplan et al. 2020). This debate between KBAI and GenAI continues (Goel 2021). Nonetheless, it is noteworthy that both KBAI and GenAI focus almost exclusively on the cognitive aspects of intelligence. When AI agents built on either paradigm do address social interactions, it is typically is on top of a full cognitive system, often to show that the cognitive system can handle social interactions, and not as a building block for developing or updating the cognitive system.

In contrast, we have posited that the development of human-level AI requires advances in computational understanding of the social aspects of intelligence in addition to the cognitive aspects. The development of the cognitive system and social interactions is bidirectional and mutually reinforcing. On one hand, the mental representations and mechanisms of reasoning enable social interactions, and on the other, social interactions afford the construction of the mental representations

and reasoning mechanisms. From this perspective, given that MToM is the basis of human-human communication, cooperation, and collaboration, computational modeling of MToM becomes essential for developing human-level AI.

We have also postulated that an AI agent's metacognition over its theory of its own mind may be a key for the construction of MToM between a human and an AI. As Kim et al. (2026) illustrate, this postulate gives computational form to Wang & Goel's (2022) three-stage process account for construction of MToM between a human and an AI. This suggests that the Symbolic/Knowledge-Based paradigm of AI may have much to contribute to the development of human-level AI viewed as a socio-cognitive system.

Acknowledgements: This research has been supported by United State National Science Foundation's Grants #2112532 and #2247790 to the National AI Institute for Adult Learning and Online Education (AI-ALOE; aialoe.org). We are grateful to former and current members of the SAMI team in Georgia Tech's Design Intelligence Laboratory for many discussions about ToM and MToM including Jisu Kim and Lingqing Wang. We thank Brittany Harbison and Spencer Rugaber for their constructive critiques of earlier drafts of this article.

References

Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). *An Integrated Theory of the Mind*. Psychological Review, 111(4), 1036–1060.

Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. *The Journal of the Learning Sciences*, *4*(2), 167–207.

Baron-Cohen, S., Leslie, A. M., & Frith, U. (1985). *Does the autistic child have a 'theory of mind'? Cognition,* 21(1), 37–46.

Brown, A. L. (1987). Metacognition, executive control, self-regulation, and other more mysterious mechanisms. In F. E. Weinert & R. H. Kluwe (Eds.), *Metacognition, Motivation, and Understanding* (pp. 65–116). Lawrence Erlbaum Associates.

Buchanan, B. G., & Shortliffe, E. H. (Eds.). (1984). *Rule-based expert systems: The MYCIN experiments of the Stanford Heuristic Programming Project*. Addison-Wesley.

Cannon-Bowers, J. A., Salas, E., & Converse, S. (1993). Shared mental models in expert team decision making. In N. J. Castellan, Jr. (Ed.), *Individual and group decision making: Current issues* (pp. 221–246). Lawrence Erlbaum Associates.

Chandrasekaran, B., & Goel, A.K. (1988) From Numbers to Symbols to Knowledge Structures: Artificial Intelligence Perspectives on the Classification Task. *IEEE Transactions on Systems, Man, and Cybernetics* 18(3):415-424.

Conway, J. R., Catmur, C., & Bird, G. (2019). Understanding individual differences in theory of mind via representation of minds, not mental states. *Psychonomic Bulletin & Review*, *26*(3), 843–858.

Corballis, M. C. (2007). The uniqueness of human recursive thinking. *American Scientist,* 95(3), 240–248

Cox, M. T., & Raja, A. (Eds**.**) **.** (2011). *Metareasoning: Thinking about thinking*. MIT Press.

Eicher, B., Cunningham, K., Gonzales, M., & Goel, A.K. (2017). *Toward mutual theory of mind as a foundation for co-creation.* Presented at the International Conference on Computational Creativity (ICCC) Workshop on Co-Creation.

Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. *Human Factors*, 37(1), 32–64.
Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*, *34*(10), 906–911.
Goel, A.K. (2021) Looking Back, Looking Ahead: Symbolic vs. Connectionist AI. *AI Magazine.* December 2021, pp. 83-85.
Goel, A. K., & Rugaber, S. (2017). Gaia: A CAD-like environment for designing game-playing agents. *IEEE Intelligent Systems*, *32*(3), 60–67.
Goffman, E. (1959). *The Presentation of Self in Everyday Life*. Anchor Books.
Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.
Grice, H. P. (1989). *Studies in the Way of Words*. Harvard University Press.
Kakar, S., Basappa, R., Camacho, I., Griswold, C., Houk, A., Leung, C., Tekman, M., Westervelt, P., Wang, Q., & Goel, A.K. (2024) SAMI: An AI Actor for Fostering Social Interactions in Online Classrooms. In Proceedings of the *20th International Conference on Intelligent Tutoring Systems*, Spring 2024.
Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., & Amodei, D. (2020). *Scaling laws for neural language models*. arXiv.
Kim, J., Islam, M., & Goel, A.K. (2026) A Metacognitive Architecture for Correcting LLM Errors in AI Agents. In Proceedings of the 40th AAAI Conference on AI, pp. 40272-40278, Singapore, January 2026.
Laird, J. (2012) *The Soar Cognitive Architecture*. MIT Press.
LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444.
Lenat, D. B., & Guha, R. V. (1990). *Building large knowledge-based systems: Representation and inference in the Cyc project*. Addison-Wesley.
McGuinness, D. L., & van Harmelen, F. (Eds.). (2004). *OWL Web Ontology Language Overview* (W3C Recommendation). World Wide Web Consortium.
McTear, M. (2020). *Conversational AI: Dialogue systems, conversational agents, and chatbots*. Morgan & Claypool Publishers.
Minsky, M. (2006). *The emotion machine: Commonsense thinking, artificial intelligence, and the future of the human mind*. Simon & Schuster.
Newell, A., & Simon, H. A. (1976). Computer science as empirical inquiry: Symbols and search. *Communications of the ACM*, *19*(3), 113–126
Premack, D., & Woodruff, G. (1978). Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences,* 1(4), 515–526
Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 10684–10695).
Rumelhart, D. E., McClelland, J. L., & the PDP Research Group. (1986). *Parallel distributed processing: Explorations in the microstructure of cognition* (Vols. 1–2). MIT Press.
Russell, S. J., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
Schegloff, E. A., Jefferson, G., & Sacks, H. (1977). The preference for self-correction in the organization of repair in conversation. *Language*, 53(2), 361–382.
Strauss, S., & Ziv, M. (2012). Teaching as a natural cognitive ability: Implications for education. *Mind, Brain, and Education*, *6*(4), 186–196.
Tomasello, M. (2008). *Origins of Human Communication*. MIT Press.
Tomasello, M. (2014). *A Natural History of Human Thinking*. Harvard University Press

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems* (Vol. 30, pp. 5998–6008)

Vygotsky, L. S. (1978). *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press.

Wang, L., Gao, Y., Anyi, C., Goel, A.K. (2026) Futuring Social Assemblages: How Enmeshing AIs into Social Life Challenges the Individual and the Interpersonal. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems,* Barcelona, April 2026.

Wang, Q., & Goel, A. K. (2021). Towards mutual theory of mind in human-AI interaction: How language reflects what students perceive about a virtual teaching assistant. *In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems,* 1–14.

Wang, Q., & Goel, A. K. (2022). Mutual Theory of Mind for Human-AI Communication. In *Proceedings of the IJCAI 2022 Workshop on Communications in Computer-Human Interactions*; available as arXiv:2210.03842.

Wang, Q., Jing, S., Camacho, I., Goel, A.K. (2020) Jill Watson SA: Design and Evaluation of a Virtual Agent to Build Communities Among Online Learners. In Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA'20). ACM, Honolulu, HI, USA.

Wang, Q., Walsh, S., Si, M., Kephart, J., Weisz, J., & Goel, A.K. (2024) Theory of Mind in Human-AI Interaction. In Extended Abstracts of the *2024 CHI Conference on Human Factors in Computing Systems (CHI EA'24)*. ACM, Honolulu, HI, USA.

Wellman, H. M., Cross, D., & Watson, J. (2001). Meta-analysis of theory-of-mind development: The truth about false belief. *Child Development, 72(3), 655–684.*

Wójcik, J., Thajchayapong, P., Tai, H., Nandan, V., Harbison, B., & Goel, A.K. (2025) Measuring the impact of AI social agents on belonging in learning communities. In *Proceedings of the 17th International Conference on Education and New Learning Technologies (EDULEARN25)*, pp. 3497-3506. IATED.